

POPE AFB, NC, USA

WMO: 723030

Lat: 35.174N

Lon: 79.009W

Elev: 67

StdP: 100.53

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13714

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.4	-4.2	-16.0	0.9	-1.6	-12.8	1.2	1.2	10.4	12.9	8.8	12.4	1.7	270	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	36.0	24.5	34.6	24.2	33.1	23.9	26.6	31.6	25.8	31.0	25.3	30.6	3.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	20.6	28.2	24.6	19.8	27.8	23.9	18.9	27.4	83.0	31.8	80.0	31.1	77.5	30.7	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.6	7.5	6.6	DB	-10.6	37.7	2.7	1.5	-12.6	38.8	-14.2	39.6	-15.7	40.5	-17.7	41.5	(4)
			WB	-11.3	27.7	2.7	1.3	-13.2	28.6	-14.8	29.3	-16.3	30.0	-18.2	30.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	6.0	8.1	11.9	17.4	21.9	25.9	27.3	26.6	23.3	17.5	11.5	8.1	(6)
(7)		DBStd	8.68	5.88	5.39	5.32	4.68	3.84	2.93	2.26	2.52	3.11	4.33	4.77	5.21	(7)
(8)		HDD10.0	429	152	92	42	3	0	0	0	0	2	38	101		(8)
(9)		HDD18.3	1568	383	288	209	72	13	1	0	3	69	212	319		(9)
(10)		CDD10.0	3049	28	39	101	226	370	478	537	513	399	233	82	42	(10)
(11)		CDD18.3	1147	1	2	10	45	125	228	279	255	152	42	6	2	(11)
(12)		CDH23.3	11444	6	17	99	464	1226	2390	2977	2607	1247	365	37	10	(12)
(13)		CDH26.7	4873	1	1	16	132	472	1115	1399	1189	455	91	2	1	(13)
(14)	Wind	WSAvg	2.7	2.7	3.0	3.2	3.4	3.0	2.8	2.7	2.4	2.5	2.4	2.3	2.5	(14)
(15)	Precipitation	PrecAvg	1154	85	73	97	79	78	120	130	126	126	75	82	76	(15)
(16)		PrecMax	1566	176	156	178	162	183	312	253	254	386	196	228	146	(16)
(17)		PrecMin	847	22	29	27	22	26	11	37	33	36	0	23	27	(17)
(18)		PrecStd	158	44	32	36	37	40	74	56	46	78	50	49	31	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.9	25.8	28.8	32.4	35.0	37.6	37.5	37.6	34.2	32.1	26.6	24.9	(19)
(20)			MCWB	18.4	17.9	17.5	20.5	22.2	24.1	25.3	24.9	23.2	22.2	19.2	18.4	(20)
(21)		2%	DB	21.1	22.8	26.1	29.9	32.8	35.3	35.8	35.3	32.5	29.0	24.1	22.1	(21)
(22)			MCWB	16.4	16.1	17.3	19.2	21.9	24.1	25.3	24.6	23.2	21.2	17.8	17.9	(22)
(23)		5%	DB	18.5	20.2	23.8	27.9	31.2	33.8	34.1	33.7	31.1	27.2	22.3	19.8	(23)
(24)			MCWB	14.5	14.6	15.9	18.4	21.4	23.6	24.9	24.5	23.0	20.1	17.1	16.9	(24)
(25)		10%	DB	16.1	17.8	21.4	26.2	29.3	32.4	32.7	32.2	29.5	25.2	20.1	17.4	(25)
(26)			MCWB	12.5	13.2	14.7	17.6	20.7	23.2	24.5	24.1	22.3	19.5	16.2	14.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.9	19.5	20.6	22.9	25.0	26.3	28.0	27.4	25.6	24.5	21.7	20.6	(27)
(28)			MCD B	22.2	23.3	24.4	26.8	30.4	32.2	33.1	31.7	30.2	27.6	24.0	22.3	(28)
(29)		2%	WB	17.7	17.9	19.0	21.6	23.6	25.6	26.9	26.4	24.9	23.3	20.0	18.9	(29)
(30)			MCD B	19.8	21.0	23.3	26.7	29.0	31.6	31.9	31.2	29.3	26.7	22.3	21.2	(30)
(31)		5%	WB	15.7	16.2	17.7	20.4	22.8	25.0	26.1	25.7	24.2	22.1	18.5	17.0	(31)
(32)			MCD B	17.9	19.5	21.3	25.2	28.1	31.0	31.3	30.6	28.2	25.4	21.1	19.0	(32)
(33)		10%	WB	12.8	13.9	16.4	19.2	22.0	24.4	25.5	25.2	23.6	20.7	16.9	15.0	(33)
(34)			MCD B	15.2	17.1	20.4	24.3	27.2	30.1	30.7	30.1	27.2	24.0	19.3	17.2	(34)
(35)	Mean Daily Temperature Range		MDBR	11.4	12.3	12.9	12.9	11.8	10.9	10.3	10.4	12.2	12.9	11.5	(35)	
(36)		5% DB	MCD BR	14.4	15.3	15.2	14.4	13.5	12.8	11.7	12.1	12.2	13.5	14.4	13.5	(36)
(37)			MCWBR	10.0	9.8	7.8	6.0	4.8	3.9	3.8	3.7	4.6	6.4	8.8	9.4	(37)
(38)			MCD BR	12.5	13.8	12.4	11.7	11.1	11.2	10.5	10.4	9.5	9.9	11.1	11.9	(38)
(39)		5% WB	MCWBR	10.0	10.3	7.9	6.0	4.6	4.0	4.0	3.7	4.1	5.7	8.3	9.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.321	0.335	0.369	0.408	0.464	0.501	0.549	0.525	0.431	0.392	0.341	0.319	(40)	
(41)		taud	2.481	2.436	2.374	2.263	2.150	2.088	1.973	2.029	2.288	2.363	2.467	2.523	(41)	
(42)		Ebn at Noon	882	908	901	877	829	796	754	765	826	830	849	863	(42)	
(43)		Edh at Noon	85	99	115	134	153	162	181	168	124	105	86	77	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.64	3.31	4.37	5.49	6.04	6.38	6.05	5.47	4.62	3.78	2.90	2.30	(44)	
(45)		RadStd	0.17	0.38	0.32	0.38	0.47	0.43	0.37	0.27	0.38	0.48	0.23	0.19	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (68 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+159	+114	(47)

Nomenclature: See separate page